

Response to Referee 3

Manuscript JFM-2026-1781

Strain-coupled one-dimensional turbulence for rapid distortion

Sharma & Kerstein

We thank the referee. Both of the fundamental objections were correct, and addressing them has changed the theory rather than only its presentation. The rigid-translation object is no longer treated as a linear-RDT benchmark; the exact projected RDT, which is broadband, is now the linear reference, and the spectral results are validated against a strained-box DNS. Section 5 no longer claims a closed single-line functional: it states the obstruction exactly, bounds the rapid term from line data, and measures the axisymmetry hypothesis instead of assuming it. Established results, assumptions and new developments are separated, the hypothesis labelled A1 is defined at first use, and the kernel equations are derived in an appendix. Section, figure, table and equation numbers below refer to the revised manuscript.

1. The linear-RDT spectral benchmark is not justified.

“the manuscript treats rigid translation of the one-dimensional spectrum as an exact linear-RDT benchmark, although this is neither derived from the full RDT equations nor demonstrated by a full spectral RDT calculation. . . . In Equation (2.14), the Fourier velocity amplitudes depend on the full three-dimensional wavevector. The manuscript provides no derivation showing that the projected spectrum must undergo a rigid translation.”

Response. We accept this in full and have removed the incorrect identification. The $d\kappa_2/dt = -A_{22}\kappa_2$ relation is now presented for what it is—the kinematics of the ODT domain dilatation (§ 2.5, equation (2.20))—and the eddy-free run that follows it is labelled the model’s *kinematic reference*, not linear RDT (§ 4.1, figure 7). The linear reference against which the line model must be judged is now the exact projected RDT, and it is computed, not asserted: § 4.2 and figure 8 give the closed-form Cauchy solution for an isotropic initial spectrum, integrated over the transverse wavenumber plane, and show that the exactly distorted, line-projected component spectra are *not* a rigid translation—the shape ratio to a rigid translation departs from unity and the components split. The manuscript now states explicitly that the reference already differs from the model’s own kinematics at the linear level. The consequences the referee flags are correspondingly revised: the claim of “suppressed spectral migration” relative to a rigid law is replaced by a measured comparison against exact RDT and DNS, and the spectral results are validated against the strained-box DNS of § 4.6 (figure 11, table 1), with the distance-from-linear-theory envelope of § 4.6.2 (figure 12) as the quantitative statement. The referee is also right that the § 4.1 (formerly Reynolds-stress-level) tests do not establish a spectral result; that is now the explicit reason the strained-box DNS was performed (§ 4.6, opening paragraph).

2. The claimed single-line closure of the rapid pressure–strain term is not closed.

“Eq. (5.7) still requires the two-variable spectral functions $a(k_2, k_\perp)$ and $c(k_2, k_\perp)$ and integration over the unresolved transverse wavenumber $k_\perp$ the paper appears to derive only an axisymmetric 2D representation, not a single-line closure. Moreover, this spectral formulation is not used in the simulations ...”

Response. The referee’s diagnosis is correct and is now the stated position of the section, not a claim it contends with. Section 5.3 carries out the axisymmetric collapse and then states its status precisely: under the axisymmetry hypothesis the rapid term is a linear functional of the two scalars $a(\kappa_2, \kappa_\perp)$ and $c(\kappa_2, \kappa_\perp)$ on the resolved half-plane (equation (5.6) with kernels (5.7)–(5.9))—“a two-dimensional representation, not yet a single-line closure, since the line determines only the $\kappa_\perp$ -integrals of fixed combinations of a and c and not the functions themselves.” The infinite-dimensional null space of the map from line data to the rapid term is stated exactly (§5.2). What the line *does* determine is then made quantitative: because the rapid term is linear in (a, c) , its extreme values over everything consistent with the line spectra, solenoidality and realisability are the solution of a family of linear programs, giving sharp bounds $[\Pi_{nn}^-, \Pi_{nn}^+]$ and a tabulated kinematic indeterminacy (§5.4, table 2; forward map and linear programs in Appendices A.2 and A.4). The representation also yields two falsifiers the line reports for itself ($\phi_{11} = \phi_{33}$ and vanishing cross-spectra), so the hypothesis is monitored rather than assumed. On the referee’s second sentence: we agree, and now say plainly that the spectral formulation is not exercised in the simulations—the reported results use the moment closure to which equation (5.6) reduces in the isotropic limit (§5.7)—and we identify the wavenumber-dependent kernel as the missing ingredient whose implementation is future work (§6.4). The section’s claim is thus reduced to what is demonstrated: an exact obstruction, sharp bounds from line data, a measured hypothesis, and exactness for axisymmetric distortion.

3. The theoretical presentation is not sufficiently clear or self-contained.

“The manuscript frequently mixes established RDT results, existing closures, assumptions, and new developments without clearly separating them. In Section 5, ‘A1’ is repeatedly used without formal definition, while Eqs. (5.8)–(5.10), described as a central result, appear essentially without derivation.”

Response. The presentation has been reorganised to separate the four categories. Established RDT results (the exact rapid pressure–strain integral and its spectral form) are attributed to the classical sources and Sagaut & Cambon (2018) and presented as standard (§§2.4.1, 5.1); the existing algebraic closures (Rotta, IP, LRR) are identified as such and confined to §2.4.2; the assumptions are stated as labelled hypotheses; and the new developments (the obstruction, the bounds, the measured hypothesis, the kernel reduction) are collected in Section 5. The hypothesis the referee refers to, A1, is now given a formal definition at first use—“Assumption A1 (axisymmetry of the undistorted spectrum about the line)” in §5.3—with its justification for the present configuration and a subsection (§5.5, table 3, figure 13) that measures the error it incurs under plane strain rather than presuming it. The kernel equations the referee cites (now equations (5.7)–(5.9)) are derived in Appendix A.3 from the azimuthal average of the exact projection weights: g_2 is worked through in full and g_1, g_3 follow by the same steps, and the two exact constraints that verify the reduction—the pointwise trace condition $g_1 + g_2 + g_3 = 0$ (equation (5.10)) and the isotropic ratio $+\frac{1}{5} : -\frac{1}{5} : 0$ (equation (5.11))—are stated and checked in the main text. We would welcome the additional issues the referee chose not to enumerate and will address them in a further revision if the paper is reconsidered.